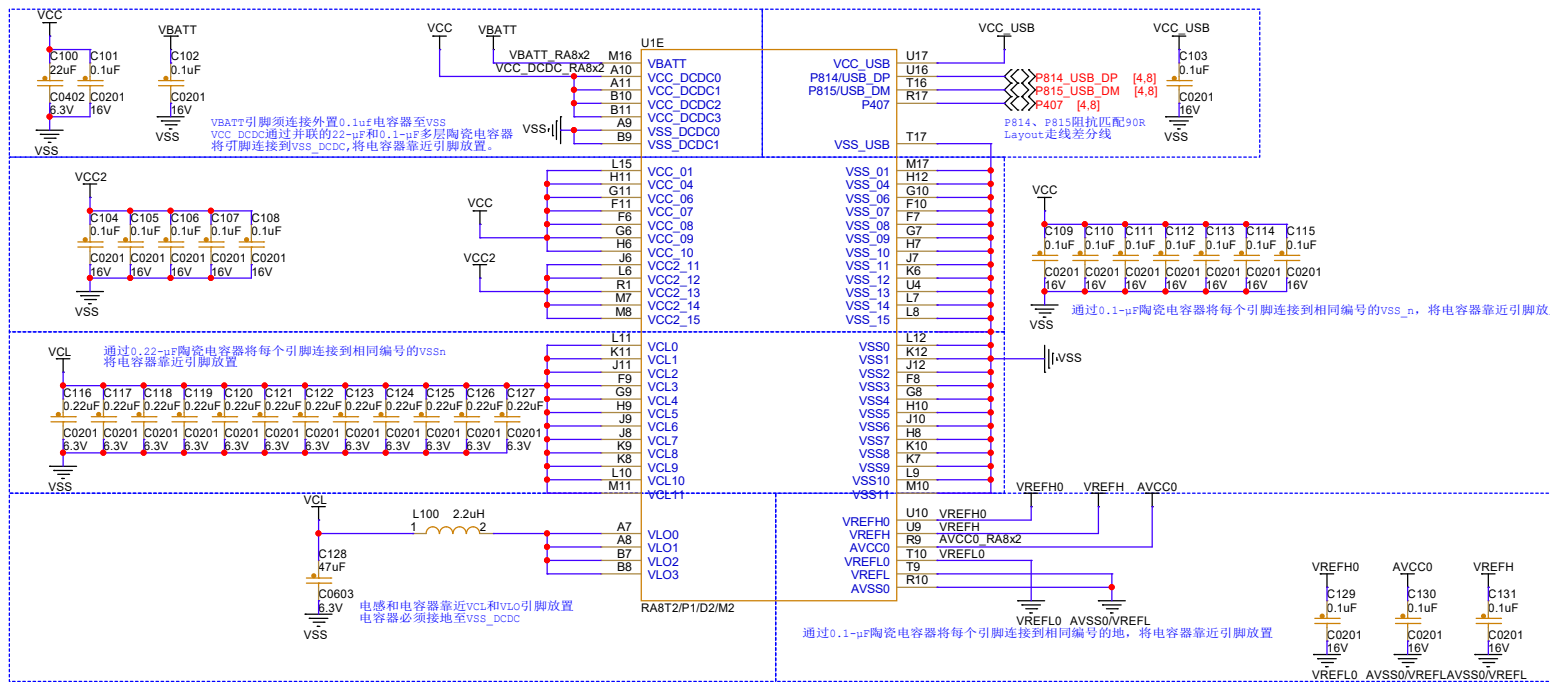




www.gitee.com/r
amcu/cpk_exam
ples

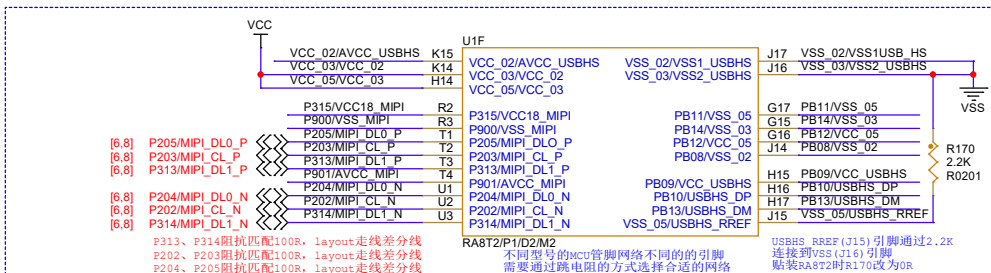


www.github.com
/renesas/cpk_ex
amples



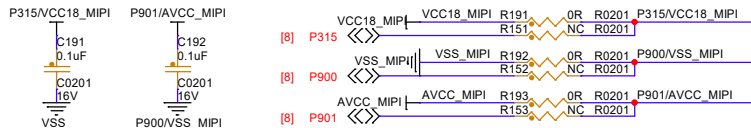
核心板省功能阻容设置

核心板芯片	R7KA8T2L	R7KA8P1K
C168-169	0.1uF	NC
C170	2.2uF	2.2uF
C171-C174	NC	0.1uF
C191-C192	NC	0.1uF
R145-R150	NC	NC
R151-R168	0R	NC
R170	0R	2.20K 1%
L171	NC	120R 100MHz
R172-R180	NC	0R
R191-R193	NC	0R



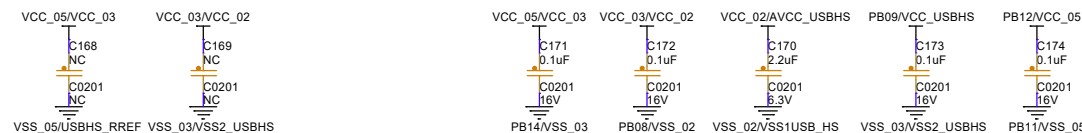
MIPI部分

连接电源的器件靠近MCU摆放，连接信号的器件在MCU-BTB间任意摆放

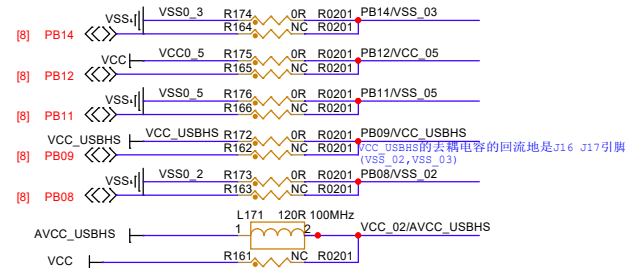


电阻选装

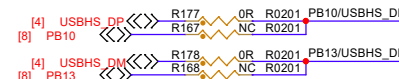
由于RA8P1/D2/M2和RA8T2的部分管脚网络不同，通过选装阻容的方式，应对不同型号的IC贴装。默认贴装RA8P1/D2/M2
连接电源的器件靠近MCU摆放，连接信号的器件在MCU-BTB间任意摆放



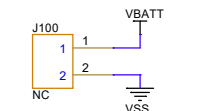
与电源相关的引脚，MCU信号出来后立即用电阻隔离



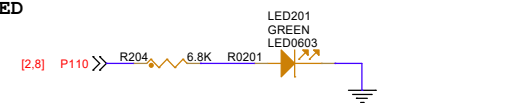
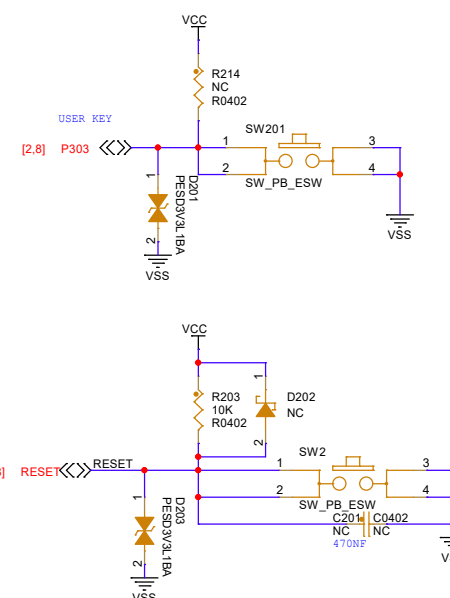
差分线走到J1(BTB)附近再用电阻隔离。



VBATT通孔接口



瑞萨电子www.renesas.com 东莞野火电子技术有限公司			
Title CPKCOR-RA8P1核心板			
Size A3	Document Number MCU 电源	Rev V1.0	
Date: Saturday, September 12, 2026	Sheet 1	of 10	



The diagram illustrates the pin connections for the RA4M2 microcontroller. It is divided into two main sections: the top section for pins 1-10 and 19-21, and the bottom section for pins 11-18.

Top Section:

- Pins 1, 2, 3, 4, 5, 6, 7, 8, 9, 10 are connected to VCC or GND.
- Pins 19, 20, 21 are connected to VSYS_3V3.
- External signals J-LINK and RA4M2_P409_CTS [3] are shown.

Bottom Section:

- Pins 11, 12, 13, 14, 15, 16, 17, 18 are connected to VCC or GND.
- Pins 19, 20, 21 are connected to VSYS_3V3.
- External signals J-LINK and RA4M2_P409_CTS [3] are shown.

Figure 1-10 shows the pin connection diagram for the TP201 and TP202 test points. The diagram illustrates two identical pin headers, J201 and J202, connected to a power supply (VCC) and ground (VSS). J201 pins 1-6 are connected to P400, P401, P402, P403, P404, and P405 respectively. J202 pins 1-6 are connected to P515, P516, P517, P518, P519, and P520 respectively. A legend on the right shows the connection for TP201 (P210), TP202 (P211), TP203 (P212), and TP204 (RESET). A note indicates a 1.25mm pitch test point order should not be乱 (messy).

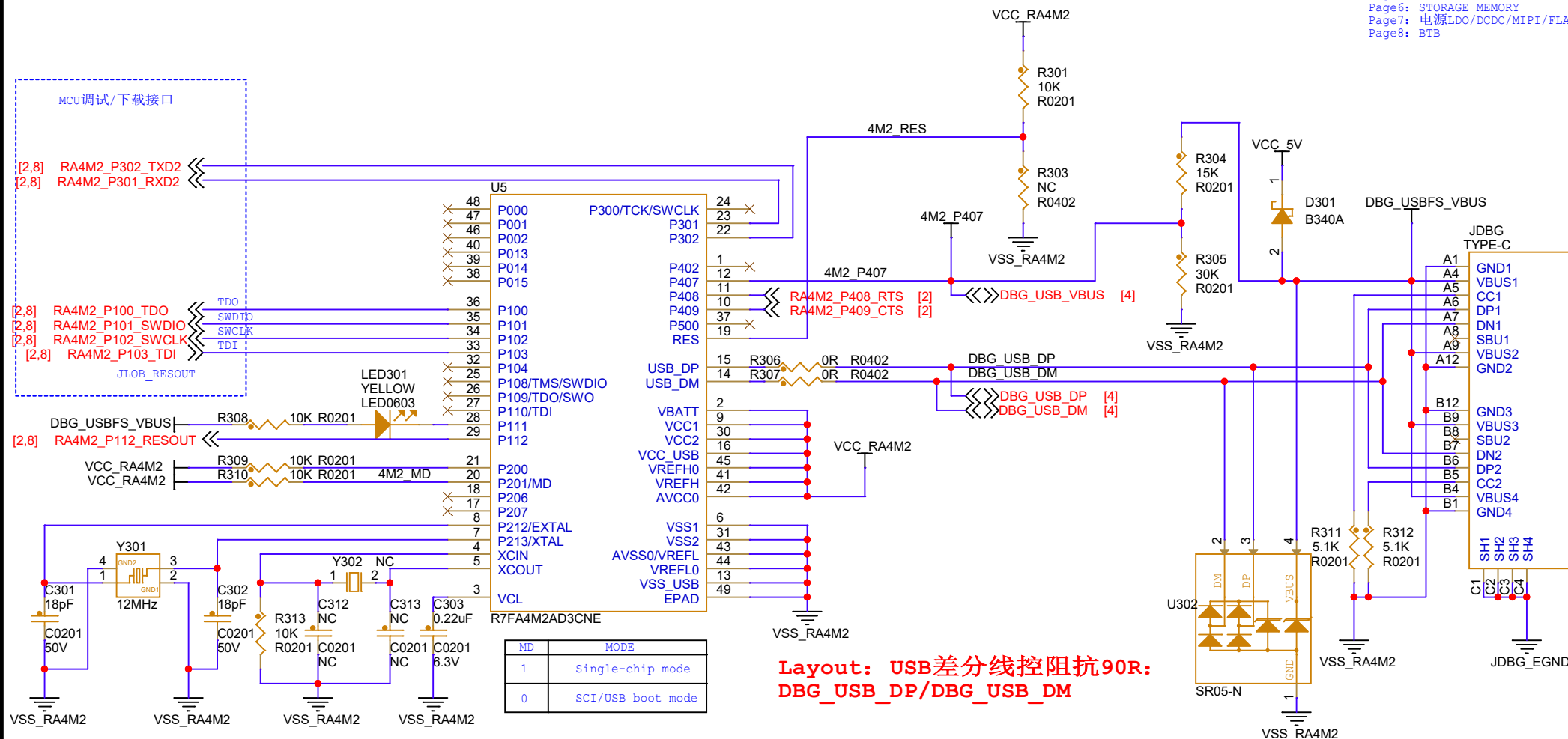
The schematic diagram illustrates the PLL circuit for the AD9548. It features two PLL blocks, Y201 and Y202. Y201 is a 24MHz PLL with inputs XTAL and XTAL, and outputs XCOUT and XGIN. Y202 is a 32.768KHz PLL with inputs XTAL and XTAL, and outputs XCOUT and XGIN. Both PLLs are connected to a 6V supply and have various capacitors (C204, C205, C206, C207, C201) connected to their pins.

1.25mm间距测试点
顺序不要打乱

瑞萨电子www.renesas.com 东莞野火电子技术有限公司		
Title CPKCOR-RA8P1核心板		
Size A3	Document Number MCU IO	Rev V1.1
Date:	Saturday, September 12, 2026	Sheet 2 of 10

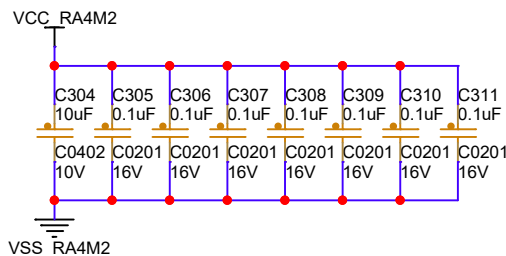
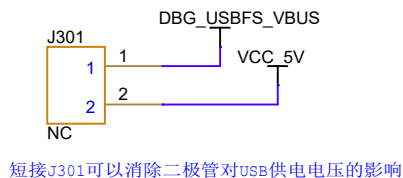
J-Link OB

Page1: MCU 电源
Page2: MCU I/O
Page3: Jlink OB
Page4: USB HS
Page5: SDRAM
Page6: STORAGE MEMORY
Page7: 电源LDO/DCDC/MIPI/FLASH
Page8: BTB



Layout: USB差分线控阻抗90R:
DBG_USB_DP/DBG_USB_DM

注意: 只能使用12MHz晶振

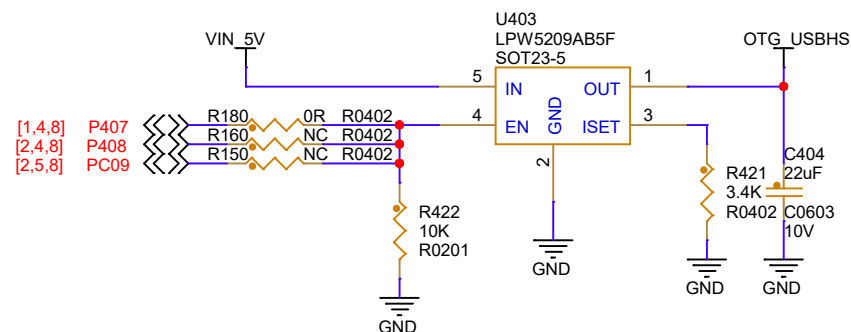
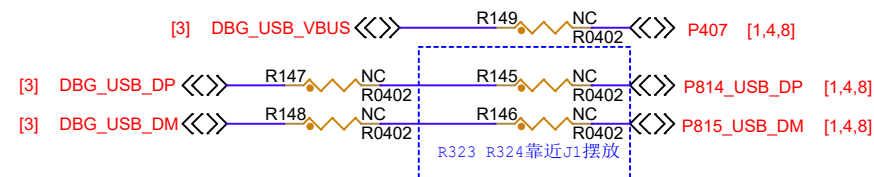


瑞萨电子www.renesas.com
东莞野火电子技术有限公司

Title CPKCOR-RA8P1核心板		
Size A4	Document Number Jlink OB	Rev V1.0
Date: Saturday, September 12, 2026	Sheet 3 of 10	

如果使用USB HS，主晶振只能选12/20/24/48MHz

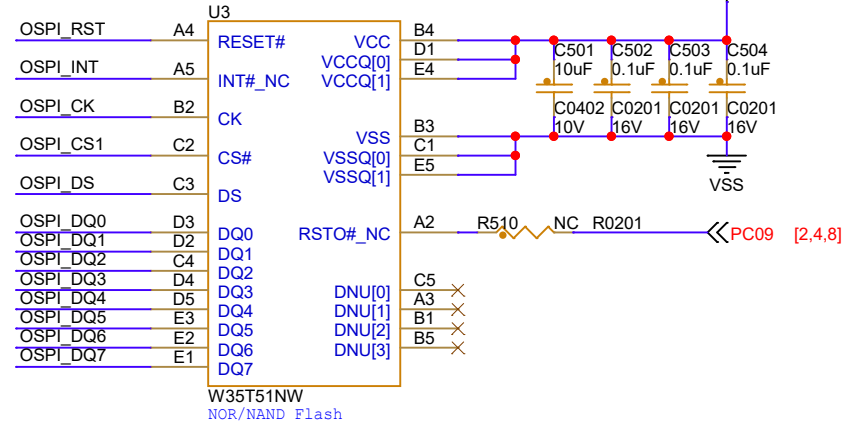
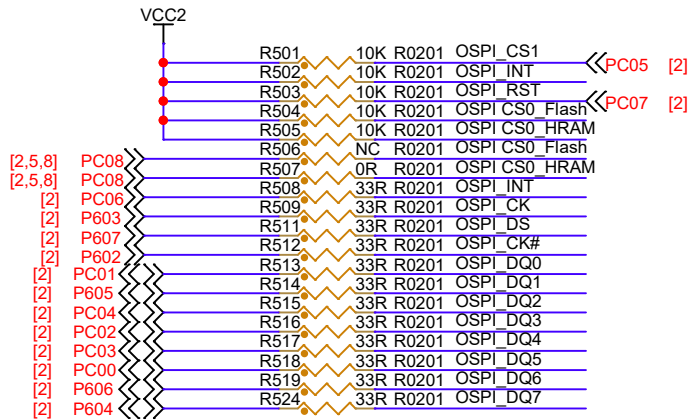
→



Title	CPKCOR-RA8P1核心板
-------	-----------------

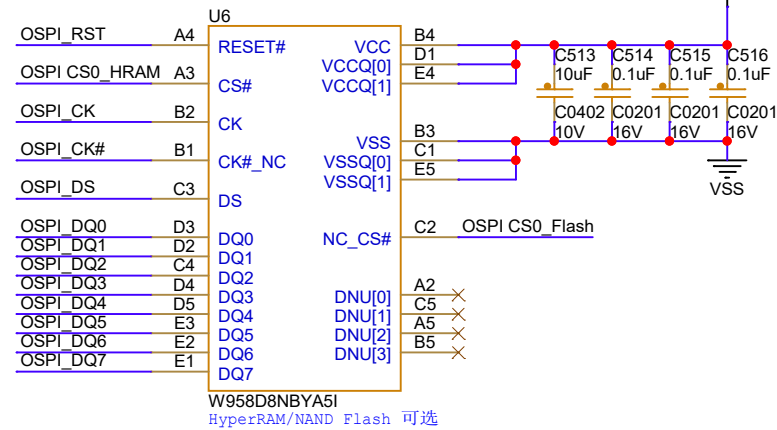
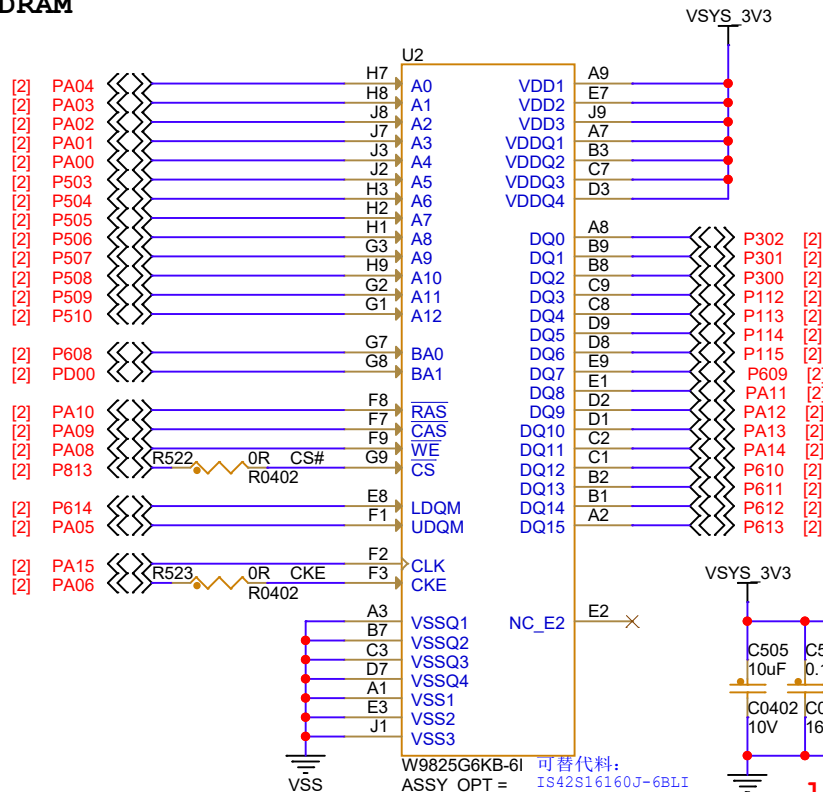
Size A4	Document Number USB HS	Rev V1.0
Date:	Saturday, September 12, 2026	Sheet 4 of 10

FLASH



Page1: MCU 电源
Page2: MCU I/O
Page3: Jlink OB
Page4: USB HS
Page5: SDRAM
Page6: STORAGE MEMORY
Page7: 电源LDO/DCDC/MIPI/FLASH
Page8: BTB

SDRAM

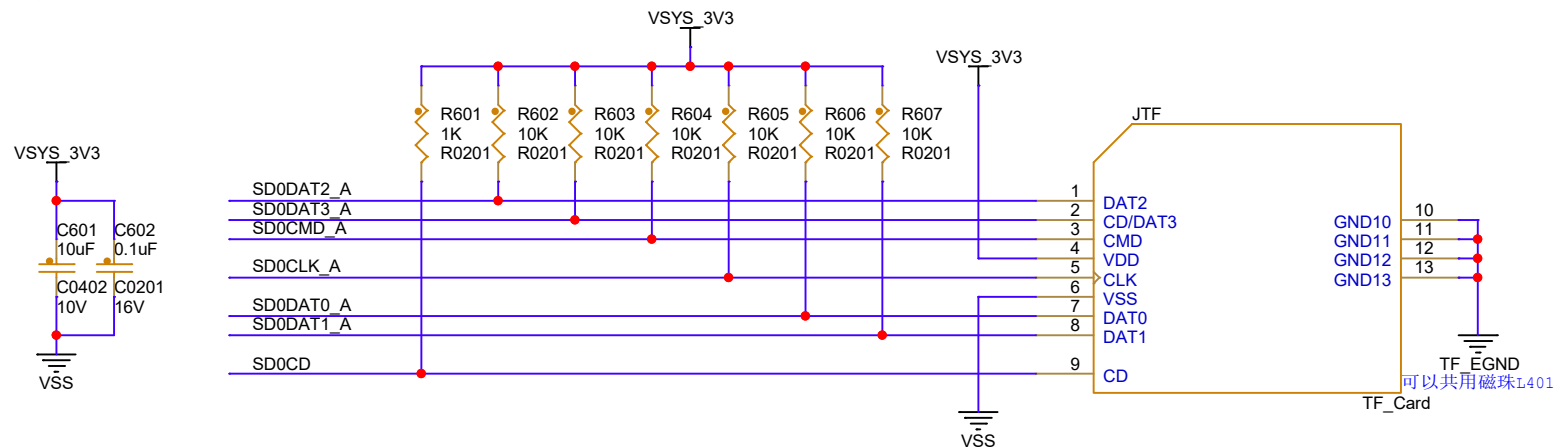


layout: SDRAM走线控阻抗50R

瑞萨电子www.renesas.com
东莞野火电子有限公司

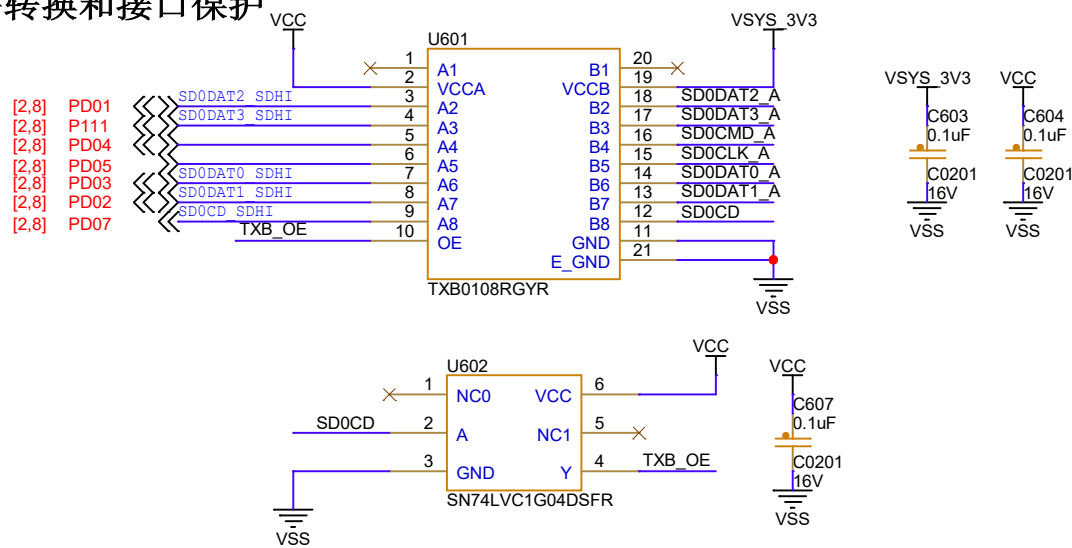
Title CPKCOR-RA8P1核心板		
Size A4	Document Number SDRAM/FLASH	Rev V1.0
Date: Saturday, September 12, 2026	Sheet 5	of 10

MicroSD卡槽

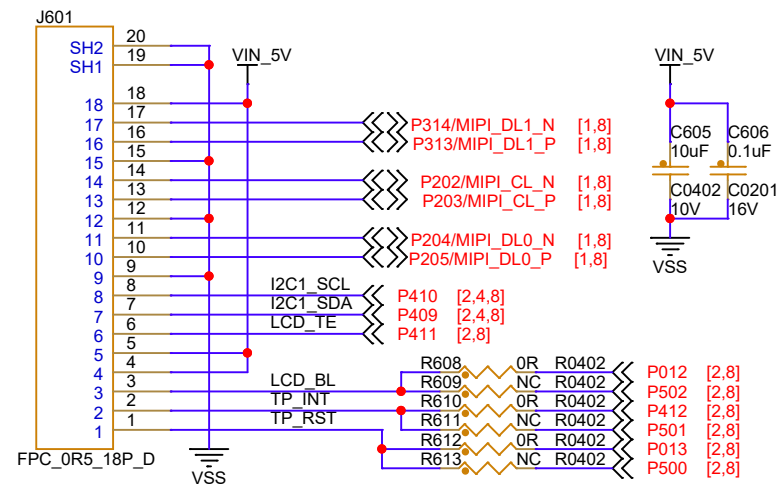


Page1: MCU 电源
Page2: MCU I/O
Page3: Jlink OB
Page4: USB HS
Page5: SDRAM
Page6: STORAGE MEMORY
Page7: 电源LDO/DCDC/MIPI/FLASH
Page8: BTB

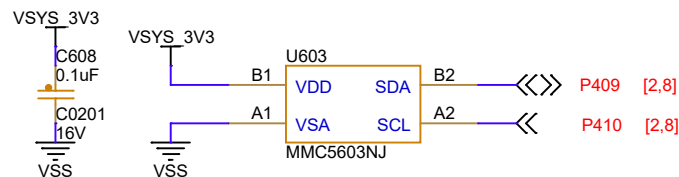
电平转换和接口保护



MIPI



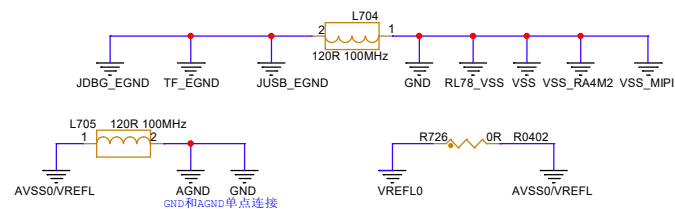
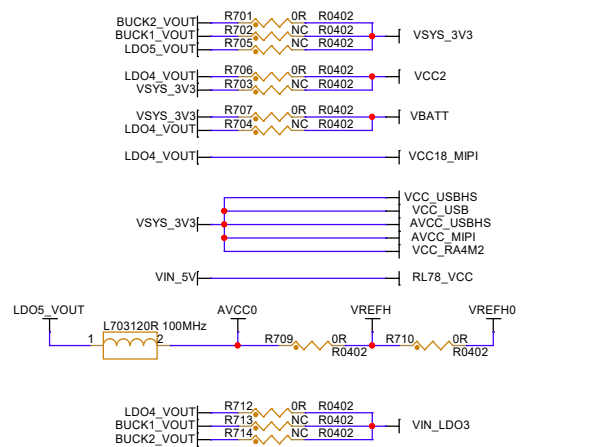
地磁传感器



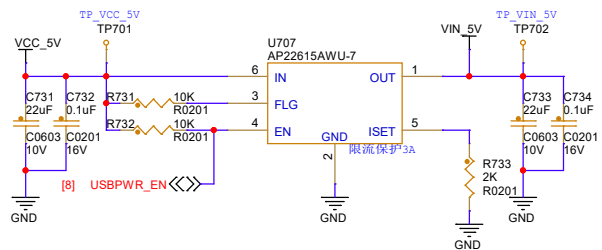
瑞萨电子 www.renesas.com
东莞野火电子技术有限公司

Title CPKCOR-RA8P1核心板			
Size A4	Document Number SD CARD/MIPI		Rev V1.0
Date:	Saturday, September 12, 2026	Sheet 6	of 10

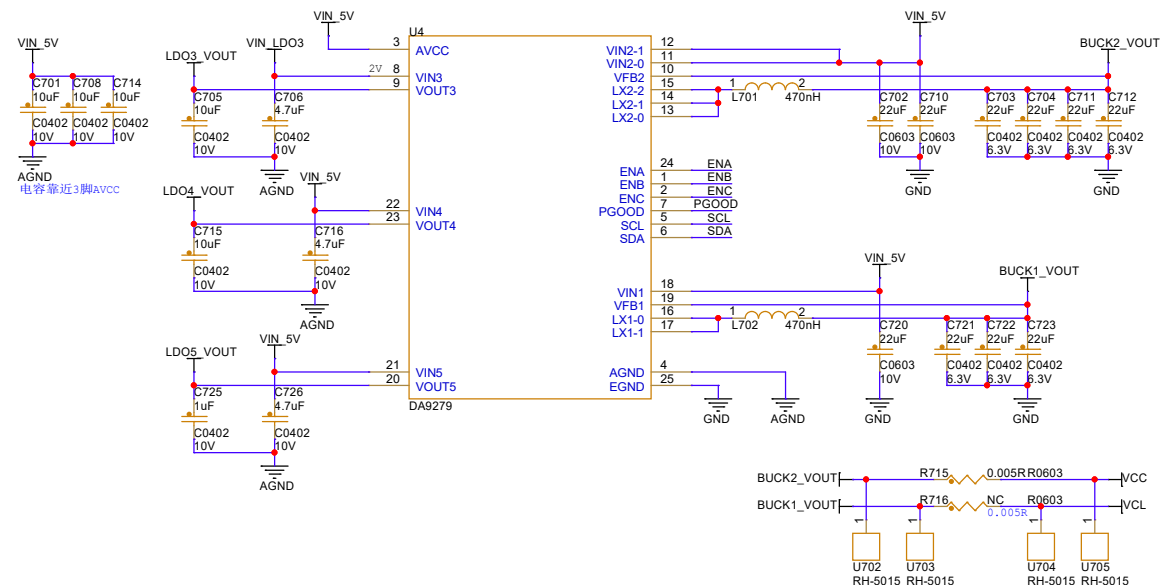
电源分割



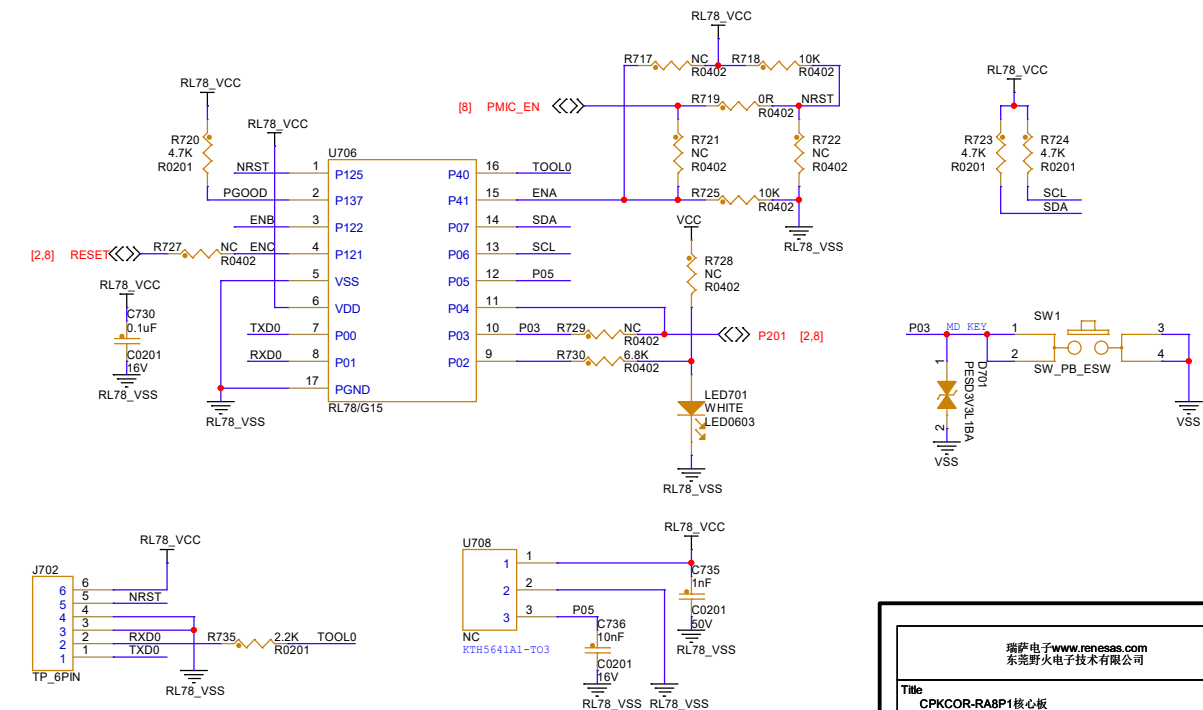
电源保护



PMIC

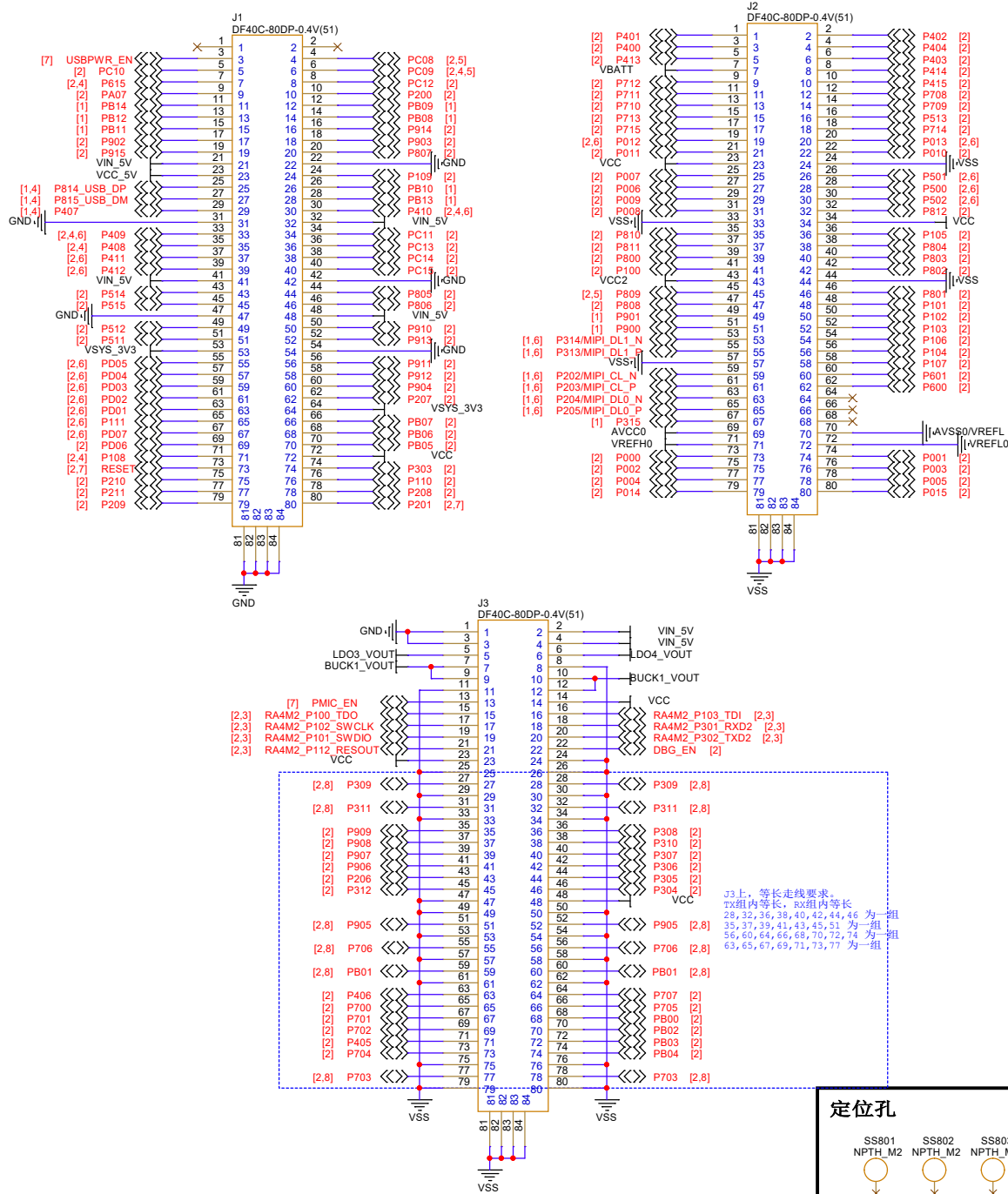


MCU



BTB座子

Page1: MCU 电源
Page2: MCU I/O
Page3: Jlink OB
Page4: USB HS
Page5: SDRAM
Page6: STORAGE MEMORY
Page7: 电源LDO/DCDC/MIFI/FLASH
Page8: BTB



瑞萨电子www.renesas.com 东莞野火电子技术有限公司		
Title CPKCOR-RA8P1核心板		
Size A3	Document Number BTB	Rev V1.0
Date: Saturday, September 12, 2026	Sheet 8	of 10

历史版本

版本号	日期	设计	描述
V1.0	2026-06-08	llm	初版发行

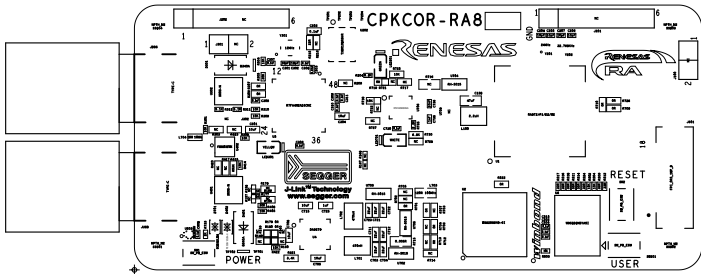
瑞萨电子www.renesas.com 东莞野火电子技术有限公司		
Title CPKCOR-RA8P1核心板		
Size A4	Document Number 历史版本	Rev V1.0
Date: Saturday, September 12, 2026	Sheet 9	of 10

CPKCOR-RA8P1核心板_原理图_V1.0

目录

Page 1	MCU电源
Page 2	MCU I/O
Page 3	Jlink OB
Page 4	USB HS
Page 5	SDRAM/FLASH
Page 6	SD CARD/MIPI
Page 7	PMIC
Page 8	BTB
Page 9	历史版本
Page 10	目录
Page 11	
Page 12	
Page 13	
Page 14	

ART FILM - ADT



ART FILM - ADT

CPKCOR-RA8P1 Power Supply Diagram

